

Overlooked Aspects of Data Governance: Workflow Framework For Enterprise Data Deduplication

Otmane Azeroual[‡], Anastasija Nikiforova^{§*}, Kewei Sha[‡]

[‡] German Centre for Higher Education Research and Science Studies (DZHW), Berlin, Germany

Email: azeroual@dzhw.eu, 0000-0002-5225-389X

[§] Faculty of Science and Technology, Institute of Computer Science, Chair of Software Engineering, University of Tartu, Tartu, Estonia,

Email: anastasija.nikiforova@ut.ee, 0000-0002-0532-3488

* Task Force “FAIR Metrics and Data Quality”, European Open Science Cloud, Brussels, Belgium

Email: nikiforova.anastasija@gmail.com, 0000-0002-0532-3488

[‡] College of Science and Engineering University of Houston Clear Lake, Houston, USA

Email: sha@uhcl.edu, 0000-0002-3750-163X

Abstract—Data quality in companies is decisive and critical to the benefits their products and services can provide. However, in heterogeneous IT infrastructures where, e.g., different applications for Enterprise Resource Planning (ERP), Customer Relationship Management (CRM), product management, manufacturing, and marketing are used, duplicates, e.g., multiple entries for the same customer or product in a database or information system, occur. There can be several reasons for this, but the result of non-unique or duplicate records is a degraded data quality. This ultimately leads to poorer, inefficient, and inaccurate data-driven decisions. For this reason, in this paper, we develop a conceptual data governance framework for effective and efficient management of duplicate data, and improvement of data accuracy and consistency in large data ecosystems. We present methods and recommendations for companies to deal with duplicate data in a meaningful way.

Keywords— *business data ecosystem, conceptual framework, data economy, data ecosystem, data governance, data management, data quality, deduplication, duplicate, duplication, uniqueness.*

I. INTRODUCTION

In the context of the data economy, which is characterized by a global ecosystem of many digitally connected actors / entities / organizations, data is considered a critical (business) asset [1]. However, many organizations are still struggling and even failing to combine a large number of internal and external data flows, assign appropriate responsibilities and determine significance and relevance to business processes to these data sources, and ensure sufficient data quality [2]. It is also believed that 80% of a data scientist’s time is spent simply searching, cleansing / cleaning, and organizing data, and only 20% - to perform analysis [3,4]. According to Total Data Quality Management (TDQM), “1-10-100” rule applies to data quality, i.e., one dollar spent on prevention saves 10 dollars on appraisal and 100 dollars on failure costs [5]. According to [6], 19% of businesses lost their customers due to the use of inaccurate, incomplete data in 2019, with losses exacerbated in industries where customers have a high lifetime value, while the 2020 edition of “Magic Quadrant for Data Quality Solutions” found that organizations estimate the average cost of poor data quality at more than \$12 million per year [7]. Moreover, according to [6], 42% of companies struggle with inaccurate data, and 43% of them have experienced the failure of some data-driven projects.

The problem of **data duplication**, in particular, has become problematic due to the growing volume of data, incl.

due to the adoption of cloud technologies, use of multiple different sources [8], the proliferation of connected personal and work devices in homes, stores, offices and supply chains [9]. Data contained in Customer Relationship Management (CRM) or Enterprise Resource Planning (ERP) systems are particularly prone to duplication [10]. Data duplication can occur due to different notions, different data formats, typos, and different abbreviations for entries such as titles for people, street names for addresses etc. Data duplication as one of the major data quality issues (also known as uniqueness) is a serious issue affecting company image, decision-making, and other data-driven activities such as service personalisation in terms of both their accuracy, trustworthiness and reliability, user acceptance / adoption and satisfaction, customer service, risk management, crisis management, as well as resource management (time, human, and fiscal). At the same time, it is known that the amount of data that companies collect is growing exponentially, i.e., the volume of data is constantly increasing, making it difficult to effectively manage them [11,12]. Consequently, organizations are affected by / suffer from inaccurate analysis, poor, distorted or skewed decisions, distorted insights provided by Business Intelligence (BI) or machine learning (ML) algorithms, models, forecasts, and simulations, where the data form the input, wasted resources, and employees, who are less likely trust the data and associated applications [13]. Thus, both *ex-ante* and *ex-post* deduplication mechanisms are critical in this context to ensure sufficient data quality and are usually integrated into a broader data governance approach [14].

Data governance (sometimes also referred to as data management) is usually based on specific (company) framework conditions that set guidelines or recommendations for the company [15,16]. Such a framework concerns the use of method structures relevant to the quality of the data used in the organization [17]. Companies can use it to determine *who can use what data, in what situations, and how, i.e., in what scenarios* [18]. The implementation of data governance can be greatly simplified with a conceptual framework [15]. Proper data governance frameworks are powerful mechanisms to help businesses become more organized and focused. They provide a structure for the data that an organization collects and guidelines for managing that data. Some data governance frameworks focus on specific areas such as data analytics, data security, or data life cycle [19-21]. However, to the best of our knowledge, there is no data governance framework for managing duplicate data in large data ecosystems, i.e., effectively, and efficiently identifying, and eliminating them. Practice also shows that many companies face challenges in

this respect in both North America, South and Latin America, Europe, Middle East and Africa, East-Asia, with the Americas being more advanced in this respect compared to other regions. A study conducted by PricewaterhouseCoopers [9] of the 2,500 largest publicly listed companies shows that while 1/3 of companies based in North America tend to have a Chief Data Officer (CDO) and deal with the data governance wiser, this is the case for only 1/4 of the surveyed companies in Europe. Elsewhere, there are even fewer numbers, as globally over 80% of CDO work in companies in North America or Europe (about 50% in the US).

Therefore, the aim of this study is to develop a conceptual data governance framework for effective and efficient management of duplicate data in big data ecosystems. To achieve the objective, we use the Apache Spark-based framework proposed by Hildebrandt et al. [21] as a reference model. It has proved its relevance in terms of generating large and realistic test datasets for duplicate detection and can go beyond the individual elements of data quality assessment. However, while this is a promising solution, our experience with it shows that it is not suitable for all data formats and database types, including but not limited to CRM, ERP, or SAP. Thus, we use it as a reference model, which we extend by integrating methods for analysing customer data collected from all types of databases and formats in the company. We believe that a data governance framework should not only evaluate, but also provide a practical guidance on how to analyse and eliminate data duplicate data through proactive management, which can then be integrated into the organization's processes.

First, we present methods for how companies can deal meaningfully with duplicate data. Initially, we focus on data profiling using several analysis methods applicable to different types of datasets, including analysis of different types of errors, structuring, harmonizing/ reconciling, and merging of duplicate data. Second, we propose methods for reducing the number of comparisons and matching attribute values based on similarity. The presented methods are designed to detect duplicates in medium to large databases. The focus is on easy integration and duplicate detection configuration so that the solution can be easily adapted to different users in companies without domain knowledge. These methods are domain independent and can be transferred to other application contexts to evaluate the quality, structure, and content of duplicate / repetitive data. Finally, we integrate the chosen methods into the framework of Hildebrandt et al. [21]. We are also exploring some of the most common data quality tools in practice, into which we integrate this framework. After that, we test and validate the framework with a real dataset. The final refined solution provides the basis for subsequent use. It consists of detecting and visualizing duplicates, presenting the identified redundancies to the user in a user-friendly manner to enable and facilitate their further elimination. By eliminating redundancies, the quality of the data is optimized and thus improves further data-driven actions, including data analyses and decision-making.

This paper aims to support research in data management and data governance by identifying duplicate data at the enterprise level and meeting today's demands for increased connectivity / interconnectedness, data ubiquity, and multi-data sourcing. In addition, the proposed conceptual data governance framework aims to provide an overview of data

quality, accuracy and consistency to help practitioners approach data governance in a structured manner.

II. DARK DATA AND DUPLICATES

On average around the world, about 55% of an organization's data are "**dark data**"— defined as data that are unknown, undiscovered, unquantified, underutilized / underused, or completely untapped [23]. In organizations, this type of data may be considered too old to be of value, i.e., obsolete / outdated, incomplete, redundant, or of limited value because it is in a format that cannot be accessed with available systems or tools [24]. In very many cases, it is not even known whether these data exist. However, dark data can represent one of an organization's greatest untapped resources [25], especially when data are increasingly becoming an important asset. E.g., one of the biggest difficulties in developing machine learning methods is getting high-quality and up-to-date training data, including but not limited historical data.

Data quality plays a special role for dark data, which is their biggest motivation and challenge at the same time [26]. For example, in research or other fields, many institutions underestimate the problem of high-quality data availability [27]. Organizations place very high expectations on the effectiveness and performance of quality processes and artificial intelligence (AI) applications. Organizations collect an enormous amount and variety of data. As a result, quality assurance and management processes (e.g., data cleansing) and employment of AI are becoming increasingly important to better understand data and use them more effectively and efficiently. In terms of data quality, this may mean that data are inconsistent, inaccurate, incorrect, or incomplete. These aspects usually also lead to the existence of duplicates, which in turn constitute contamination / pollution.

Duplicates are common in dark data, and they greatly affect the quality of the data. In general, it can also happen that so-called duplicates exist in different types of data. This means that there are multiple representations of the same real-world object in a database or information system. The reasons for the existence of duplicates can be very diverse. In the context of a single database, for example, the causes can be incorrect records / entries due to typos or multiple entries, different input conventions, or significant changes in the affected real-world object between two. Another more common and frequent source of duplicates is the integration of multiple databases, information systems, or repositories, where e.g., different data schemata, naming structures, conventions or input requirements can lead to different representations of the real-world object. Effective and efficient identification of such duplicates is a complex task and falls within the domain of a duplicate detection [28].

In general, it can be speculated that dark data and duplicates, given their nature, need to be considered from different perspectives, where not only technological solutions are needed that would reveal poor- / low-quality data, but also organizational and cultural changes related to data management and governance within the organization are needed. These are perspectives that we consider further, proposing respective solutions.

III. METHODS AND WORKFLOW FRAMEWORK FOR DUPLICATE DETECTION

Recognizing the need for duplicates management, we present a set of expected requirements and a list of practices that can be integrated into our data governance framework.

A. Requirements

To do be consistent with the motivation and intended purpose of duplicates management, the planned procedure must meet a number of criteria. The most important aspects identified in [21] form the following requirements:

- **efficiency & scalability:** the primary requirement for a method is that it should be able generate very large test datasets in an acceptable run time. Thus, the process must at least be efficient enough or scale horizontally. At this point, both aspects are formulated as requirements: **(R1) the highest possible efficiency and (R2) scalability**;
- **schema and data type independence:** the method must be able to obtain / derive test datasets from any existing relational datasets. To do this, **(R3) it must be able to handle different schemas and data types**;
- **realistic errors:** the input is assumed to be a dataset that is as clean as possible, thus **(R4) the method is also expected to be “responsible” for injecting errors**. According to the garbage-in/garbage-out principle, only test datasets with realistic errors can lead to meaningful evaluations. Thus, in order to be able to use the generated test datasets in duplicate detection in a meaningful way, **(R4) the injected errors should match as closely as possible the errors of the real datasets in respective domain**;
- **flexible configurability:** the method should **(R5) allow generating different test datasets** with different properties from an input dataset, depending on the configuration. This includes, for example, parameters such as the number of tuples, the proportion of duplicates, the degree of contamination / pollution, or the type of errors contained. On the other hand, **(R6) the required configuration effort for the user should be as small as possible** to make this tool easier to use and enable inexperienced users. It is important to reconcile these opposites.

To meet the above key requirements, a concept was developed that embeds a number of methods into a workflow framework. Each method serves as an aid to the implementation of certain requirements. These methods are explained below, followed by the details of the entire workflow framework.

B. Methods

The task of identifying duplicates is usually solved in a process-driven way. The duplicate detection process consists of several methods and can be designed very differently, but, following relatively similar structure, which is provided below and is used in the created workflow framework:

- 1) **Data Profiling** describes automated data analysis using various analysis methods and techniques [29].

In the framework presented, profiling of data associated with attributes is used to determine statistical characteristics of individual attributes.

These are recorded in the data profile along with the schema information of the data source;

- 2) **Data Cleaning** (also called data cleansing, data pre-processing or data preparation) is used to prepare and standardize the database [30]. Data quality can be improved through syntactic and semantic adjustments, which in turn can improve the quality of duplicate detection. Possible measures include removing unwanted characters, standardizing abbreviations or adding derived attributes;
- 3) **Table schema enrichment.** Depending on the type of data source, the level of detail of database relationship schemas may vary, i.e., contain a different amount of information. While the CSV file specifies the number and, if applicable, the names of the attributes, the MySQL table schema, for example, also contains information about the data types of the attributes or whether an attribute can contain NULL values [19].

The presented framework uses an extended / enriched table schema, which includes typical information about the structural schema of the input dataset, and also contains additional information for further processing of the data. On the one hand, this allows the procedure to be schema independent and, while it also lays the foundation for generating realistic errors and implementing flexible configurability;
- 4) **Automated pre-configuration.** With the help of reasoning procedures, a pre-configuration of the actual data generation process is automatically derived from the characteristics of the input data [19]. Certain rules or machine learning classifiers can be used for reasoning. The generated pre-configurations are intended to ensure that the required configuration effort remains as low as possible despite the large number of configuration parameters that the user can manually configure / adjust;
- 5) **Search space reduction** (indexing or blocking). Most of the methods for detecting duplicates are based on pairwise comparisons of tuples. Therefore, the search space for duplicates, and hence the complexity of detecting duplicates, is in principle quadratic to the tuple set. Thus, methods for reducing the search space are essential for the duplicate detection process to be sufficiently efficient, especially in the case of larger sets of tuples [31]. Such methods try to avoid comparisons between tuples that are very likely not duplicates, using certain criteria. When choosing the criteria used for this, there is always a compromise/ trade-off between the reduction factor or ratio and the completeness of pairs. The most well-known methods for reducing the search space are *standard blocking* and the *sorted neighbourhood method* [12];
- 6) **Attribute value matching**, i.e., values are calculated for each pair of tuples in the (reduced) search space using appropriate measures to represent the degree of similarity between the attribute values of the two tuples [32]. Various similarity measures can be used for this, e.g., edit-based (*Levenshtein distance*), sequence-based (*Jaro distance*), or token-based string metrics (*Jaccard coefficient*, *n-grams*) [32];

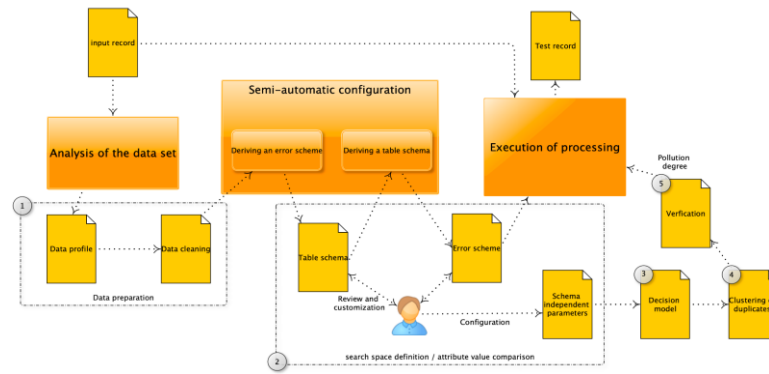


Fig.1. Workflow framework for data discovery (based on [21] and [12])

- 7) **Error model**, which is based on the so-called error schemes, where each schema represents a specific defined sequence of applying of different types of errors to data. This process is defined by flexible linking and nesting of different types of errors using meta-errors. Using probability values and weigh coefficients, such a process can be partially randomized [33]. The combination of these concepts enables flexible configuration. Error types are classified according to their scope, i.e., according to the area of the dataset in which they operate. Row errors refer to individual rows (i.e., tuples), column errors refer to individual columns (i.e., attributes), and field errors refer to individual table fields (i.e., attribute values) [33]. The scope distinction is useful for concurrent execution and distribution of error application (perhaps in partitioned data) and thus supports the requirement for efficiency and scalability. These field/area errors can also be further divided into subclasses based on the data type of the field. Different errors for numeric values, words or sequences of words. This distinction allows flexible handling of different types of data and the generation of more realistic errors [32].

So-called groups of tuples represent another dimension of the configuration of realistic error schemes. With this concept, groups of tuples can be defined as having a common group error schema / pattern. In this way, structural, syntactic, and semantic heterogeneity can be created between tuples from different groups of tuples to reflect different data sources (e.g., a database or CSV file), input sources (e.g., a form or API), survey contexts (e.g., the conclusion of contract or non-binding discussion) or points in time to simulate a survey [19]. A single group of tuples is defined using a condition in the error schema, which is defined using the table schema, such as $ID \leq 1000$ or $CITY = "Berlin"$. All tuples that satisfy this condition belong to the tuple group of this error scheme. Since individual conditions need not be mutually exclusive, a tuple can also belong to more than one tuple group.

- 8) **Classification (tuple matching or decision model)**, in which compared pairs of tuples are divided into *matching*, *non-matching* and *possibly matching* (i.e., “duplicates”, “non-duplicates” and possibly “possible duplicates”) based on their values similarities [35]. For this, probabilistic methods, distance methods, rule-based methods, and supervised and unsupervised

machine learning methods can be used. As a rule, the decision to classify a pair is made independently of other pairs of tuples [35]. This means that the results may be inconsistent and even contradictory. It may happen that the pairs $\langle t1, t2 \rangle$ and $\langle t1, t3 \rangle$ are classified as matching, while the pair $\langle t2, t3 \rangle$ is classified as non-matching, which violates the transitivity of the duplicate relation [36].

- 9) **Clustering**. After the classification is completed, a (possibly inconsistent) classification of pairs of tuples into matches, non-matches / mismatches, and possible matches takes place [38]. However, duplicate detection is defined as mapping an input relation to a cluster. In order to obtain such a globally consistent result, the tuples are collectively classified in the clustering step based on (independent) pairwise classification decisions. This means that information about duplicate relationships between different tuples is usually taken into account in order to partition the input relation into disjoint duplicate clusters [39].
- 10) **Verification**. There are various metrics to evaluate the quality of the duplicate detection process. It should be noted that duplicate clustering can always be expressed as a set of pairwise classification decisions. The usual approach is to express the quality of the process in terms of the goodness of these pairwise classification decisions. The numbers of correctly classified duplicates (true positives), tuple pairs incorrectly classified as duplicates (false positives), and duplicates not found (false negatives) are related. The most common of these measures are Precision, Recall and F-Measure [40]. Another approach measures the similarity of clustering result to *gold standard clustering*. Examples of these metrics include the *Rand Index (RI)* and the *Talbur-Wang Index (TWI)* [41].

C. Workflow Framework

A prerequisite for applying the methods presented in this paper is the existence of a clean dataset with a tabular structure that is as clean and free of duplicates as possible, and that each clean dataset can serve the same purpose. Taking such a dataset as input, a duplicate detection test dataset is generated using the workflow framework shown in Fig. 1. The workflow framework essentially consists of the following three core phases: (1) analysis of the dataset, (2) semi-automatic configuration, (3) execution of processing.

1) *Analysis of the Dataset*: A clean dataset is imported and analyzed using attribute-based data profiling and data cleaning techniques / methods. First, the existing data source schema information is determined. Then, for each attribute, a set of statistical information about the associated attribute values are collected and recorded in the data profile for the corresponding attribute. This data profile contains information about:

- **distribution of the base data types**: for each attribute value of an existing attribute, an attempt is made (regardless of other attribute values) to determine which base data type it represents. It then counts and records how often the different underlying base data types in the attribute were determined. In addition, the number of NULL values contained is recorded. The set of base data types depends on the implementation, but must cover at least four types, namely *string*, *double*, *integer*, and *Boolean*;
- **base data type in the schema**: if the schema of the input dataset specifies a data type for the attribute being examined, it will be recorded here. If no data type is specified, the most common data type found in the distribution of the basic data types is assumed to be the base data type;
- **uniqueness of attribute values**: the number of individual/ unique values (distinct value count) and their share of the total number of attribute values is calculated (distinct values ratio);
- **data type-specific information**: depending on the basic data type of the attribute under consideration, some information about the content of the attribute's values may also be of interest. For example, for *numeric* and *string* attributes, this could be:
 - **statistics for numeric attributes**: if the attribute is *numeric* (e.g., *integer*, *double*), statistical information about the attribute's values, such as min, max, average, and spread of attribute's values, is collected;
 - **statistics for string attributes**: if it is a *string* attribute, basic statistical information (same as for numeric attributes) is collected about: (1) the number of tokens per *string*, and (2) lengths of each token.

2) *Semi-automatic Configuration*: The information obtained from the analysis of the dataset can be used in this step to create a configuration of schema-specific aspects for the next step (execution of the processing process). For this purpose, pre-configurations are automatically derived from data profiles and data cleansing, which can then be examined and adjusted manually by the user:

- **Table Schema Configuration**: based on the results of the analysis step, data profiles, data cleansing, a table schema is derived, which contains:
 - **the number and names of the attributes** that correspond to those in the input dataset. If there are no names, the attributes are numbered sequentially (i.e., attribute 1, attribute 2, attribute 3, ... attribute n);

- if a data type is specified in the attribute schema, **the base data type** is derived from it. Otherwise, it is derived from the distribution of baseline data types collected in the analysis phase. When multiple base data types are included in a distribution, a specific strategy is required to select a single base data type. For example, the most specific base data type can be selected, which can represent all attribute values without any loss of information;
- in addition to the basic data type, **an abstract data type** can also be assigned to an attribute. This includes more specific concepts such as word, phrase, or text to distinguish between string types. The assignment of such abstract data types to attributes can be done based on the data profile, e.g., with the help of reasoning systems, or manually by the user;
- if the schema of the input dataset does not define **whether the attribute is nullable**, this property can be set by having NULL values in each attribute value;
- if the schema of the input dataset does not define **whether the attribute is unique**, this property can also be obtained / derived from the individuality of its attribute values (ratio of distinct values);
- Boolean value allows the user to specify whether the attribute should be considered for pollution at all;
- finally, a suitable similarity measure that can meaningfully compare this type of data value is derived from the determined information about the attribute.

The table's derived schema should be considered as a suggestion and can be inspected and modified by the user.

- **Error schema Configuration**: the error schema is then generated from the table schema. It is concluded which attributes have which error types, because not every error type is suitable for every attribute. This pre-configuration should also be interpreted as a suggestion and can be adjusted manually by the user.

In addition, some schema-independent processing aspects must be defined. They are manually configured by the user, which in principle can be done completely independently of this phase (same as the first) in terms of causality and time. The schema-independent aspects are:

- **Duplicate Cluster Configuration**: this step determines how many duplicate clusters of which cardinalities to create. This can be done in three ways:
 - **exact**, by specifying the desired numbers of duplicate clusters of certain cardinalities;
 - **absolute**, by specifying the absolute number of duplicates desired and a distribution to calculate the exact duplicate configuration of the cluster. *Example: 5,000 duplicates; normal distribution*;

	Data quality tools										Core functionality									
	Data Profiling										Data integration					Data cleansing				
	Column value frequency analysis and related statistics	Table structure analysis	Domain analysis	Cross-table redundancy analysis	Business rules documentation	Pattern library	Data pattern recognition	Drill-through analysis	Data extraction, transformation and consolidation	Metadata management	Data dissemination	Business modeler and job designer	Data access	Parsing, standardization, enrichment, and merging	Data deduplication	Record matching and identity resolution	Address verification and address validation			
Talend Open Studio	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗
Talend Open Profiler	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗
DataCleaner	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗
Pentaho Kettle	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗
Ataccama Data Quality Analyzer	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗
SQL Power Architect	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗

Fig. 2. Data quality tool analysis report (based on [42]).

- **relative**, by specifying the relative proportion of the desired duplicates to the source (or target) size and distribution to calculate the exact configuration of the cluster of duplicates. *Example: 50% of the created tuples should be duplicates; Equal distribution.*
- **Pollution level configuration**: a value that specifies how polluted the tuples in the duplicate clusters should be (approximately) is set at this stage. The degree of pollution/contamination is modelled using the similarity value. There are various ways to calculate this value. E.g., it can be the result of all average similarity values of all pairs of tuples of all duplicate clusters. The specified size depends on the data and the similarity measures used.

The entire configuration can be saved and later reloaded to establish a certain reproducibility of the generated test data records, since exact reproducibility cannot be guaranteed due to the randomized nature of the processing procedure. Obviously, metadata are also recorded, i.e., the name of the input file, the date and time.

3) *Execution of Processing*: This is the key stage where the data pollution / contamination and hence the actual generation of the test dataset is performed. It consists of:

- 1) **injection of duplicates**: according to the configuration of duplicate clusters, a tuple is randomly selected for each duplicate cluster C and duplicated $|C|-1$ times, where $|C|$ stands for the respective cluster size. The created duplicates are added to the dataset. It keeps track of which tuples belong to the same duplicate cluster;
- 2) **error injection**: a previously defined error scheme is iteratively applied to the dataset until the desired level of pollution / contamination is reached. To do this, the degree of contamination achieved is calculated between the individual iterations and compared with the desired target value from the configuration. Once this target value is reached, the loop is terminated. Due to this simple mechanism, the accuracy/precision of the approximation to the desired level of pollution depends on the pollution effect of the error scheme. The more iterations of the error scheme that pollutes the dataset, the greater the

potential difference between the result and the desired level of pollution.

It should also be noted that the error scheme represents a defined process. As a result, error applications usually have a fixed order within an iteration. Also, the error schemes of the various groups of tuples are executed one after the other according to the order in which they were defined;

- 3) **preservation**: the modified dataset is saved as a test dataset. Its gold standard is added by persisting the tuple cluster ID as an additional dataset attribute.

D. Data Quality Tools

Today, there are many data quality tools on the market. Choosing the right tool is critical to implementing the above proposed framework. Among the available tools, probably the most acceptable in the industrial settings are *Microsoft SQL Server Integration Services (SSIS)* and *Data Quality Services (DQS)* [43]. *SSIS* provides a wide range of capabilities for data migration and ETL (extract-transform-load) processes that allow users to collect data from any source and enrich it during migration and integration operations. *SSIS* is mainly used for data migration tasks. *DQS* is a knowledge-based data quality tool (a SQL Server component) designed to analyse and improve data quality. Its main functional components are: knowledge base, data reconciliation, data cleansing and data profiling. However, *DQS* has several drawbacks, including the ability to analyse only one table at a time (analysis of more than one table is only possible, if the view is first created from them), domain format limitations, a minimum data matching threshold of 80%, and high resource consumption when analysing large amounts of data [43]. At the same time, most of the data quality tools available, such as the ones from *Experian Information Solutions*, *Informatica*, *Information Builder*, *Microsoft Corporation* and *Oracle Corporation* are expensive. Recently, a number of commercial open-source data quality tools have been developed that companies use to ensure and maintain data quality or achieve other strategic goals on a daily basis. Our experience shows that in the area of data profiling and data cleansing, *DataCleaner*, *Ataccama's Data Quality Analyzer*, *Pentaho Kettle*, *SQL Power Architect*, *Talend Open Studio* and *Talend Open Profiler* provide the most needed features. Focusing more on data integration, cleaning, and enrichment methods and techniques, these tools include a set of downloadable components that allow to connect to various data sources.

To facilitate understanding in choosing the right data quality tool for this study, a comparative evaluation of the open-source tools developed in [42] was examined, the results of which are presented in Fig. 2. In particular, it evaluates open-source tools or those for which free trial versions, as well as examine practical applications of tools in data quality programs that support data management initiatives. The tools were evaluated based on performance characteristics and the type of data quality issues they deal with.

Open-source tools provide a sort of leverage by pointing out problem areas and helping to quantify the resources needed to fix the problems. When choosing any tool, there must first be a specific plan that fully covers the purpose, scope, and expected results. This is where data quality tools can make a big difference. All of the tools mentioned have a user interface that provides the user with the features and functions necessary to analyse and maintain data quality. In most cases, they are sufficient to maintain data quality at an acceptable level. Most of these tools are designed with a focus on specific features or functions and are especially useful for projects with a small budget. The results presented in Table 1 show that *DataCleaner* offers the widest range of functions in the three core functional areas and provides a quick insight of the possible causes of data quality problems. *Pentaho Kettle* and *Ataccama Data Quality Analyzer* are the next important tools (the 2nd, and the 3rd). *SQL Power Architect*, *Talend Open Studio*, and *Talend Open Profiler* offer the least number of expected features. In the view of the above, the preference in this study was given to *DataCleaner*. According to the investigation and evaluation of various open-source tools, *DataCleaner* is the most suited solution, which provides comprehensive, cost-effective, plug-and-play solution to improve data quality, offering the most support for data profiling and data quality monitoring compared to other tools.

IV. TESTING WORKFLOW FRAMEWORK FOR DUPLICATE DETECTION

To evaluate the practical relevance of the developed workflow framework for automated detection of duplicates in large volumes of data, and to explore the strengths and extensibility of the approach, we developed procedures based on the *DataCleaner* tool and applied them to the test case. Both the used dataset and the code for integrating the framework into the *DataCleaner* tool are publicly available on GitHub (<https://github.com/OtmeneAzeroualDZHW/Record-Linkage-Based-Data-Deduplication-Framework>), and for the sake of simplicity we won't discuss it here in detail, rather focusing on the user's perspective.

Fig. 3 provides an overview of how data quality issues such as duplicates can be analysed by integrating the procedure with the *DataCleaner* tool.



Fig. 3. Data quality tool analysis report (based on [42])

For the testing purposes we used the dataset in .csv format that contains a collection on vehicle data. This dataset contains 5000 records, where each vehicle is described by 250 parameters (i.e., 5000 rows, 250 columns). We were interested in identifying duplicate product names since these names are expected to be unique. When the dataset was loaded into the *DataCleaner* tool in which the above procedure was integrated, and the data were analysed, identical product names that were duplicates are quickly identified. I.e., the result returned to the user shows that the product named "Classic Cars" occurs 62 times in the dataset, "Motorcycles" – 37 times, and "Trucks and Buses" – 11 times, which can also be seen from the bar chart (aka column chart) that shows the overall distribution of all values the attribute of interest has. The user interface has a green arrow next to the numbers in the *Match Count* column, which, when clicked, provides the user with a list of duplicate entries. This makes it easy to navigate to these duplicates, examine them, including investigating their origin (root-cause analysis), and finally eliminating them at the source. Then, an extensive duplicate analysis is expected, in which the data must be fully checked, and duplicates eliminated. It is also necessary to determine their source of origin, as incorrect data have a drastic impact due to the central data store. In other words, a central data storage, database, or information system, can be accessed by multiple departments and enterprise applications, or external parties, which then reuse them. Therefore, even a small data error can permeate the entire company and business, causing subsequent errors and provoking wrong decisions or any other data-driven actions.

The more data are generated and used, the more important data governance becomes. At first glance, the design and implementation of such workflow framework seems time-consuming and complex. However, in the long run, companies with a functioning data governance strategy win, where the establishment of the above is a one-time event with only its maintenance expected. The developed workflow framework offers users a systematic approach to data quality management. Its use is an effective way to control duplicate data and other data quality issues, thereby reducing anomalies and inconsistencies resulting from uncontrolled redundancy. The framework allows users to proactively / preventively filter incoming data volumes. It also offers dynamic, configurable / customizable, portable, and automated duplicate handling. It can also be extended to offer the ability to add data quality metrics and other analysis modules.

V. DISCUSSION

Which data within a company requires the most careful control depends on business factors, as well as the nature of the data the company is dealing with. E.g., for customer service, ensuring the quality and protection of customer-related data is of particular importance. Therefore, the secure management of data as it is exchanged within an organization (or outside) is essential to ensure compliance with legal and industry-specific regulations. These requirements form the basis of a company's data governance strategy, which in turn forms the basis of the data governance framework. Various company-specific factors have a strong influence on the design of the data governance framework, which can be (1) the organizational landscape, (2) the competency landscape in the organization, (3) the competency landscape in the IT department, (4) the strategic landscape, as shown in Fig. 4.

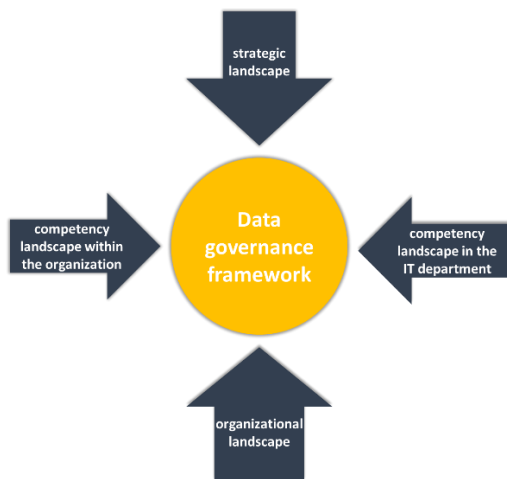


Fig. 4. Factors influencing the data governance workflow framework

Already existing and established business processes and strategies, and technical/organizational structures are essential to the implementation of the framework. Data governance, closely related to data quality management, affects the three levels of decision/impact: strategy, organization, and information system during implementation. After formulating the data quality strategy and the first package of measures, the specifications are operationalized by defining data control processes (data maintenance). This also includes assigning responsibilities and tasks to roles. The data governance framework has the following typical roles that can be responsible for the workflow framework:

- **Data Governance Committee**, whose role is to define the data governance framework based on both company-related aspects, available data artifacts with the reference to both the data and their nature, as well as the systems that deal with these data, and stakeholders and actors (referring to both internal and external) dealing with the data. This committee is also expected to oversee its further implementation across the organization.
 - **Chief Steward**, whose role is to implement the control structures and define the requirements for the implementation of the data governance framework;
 - **Business Data Steward**, whose role is to detail the data governance framework for the area of responsibility from a business perspective;
 - **Technical Data Steward**, whose role is to create technical standards and definitions for individual data elements, and to describe systems and data flows between systems from a technical point of view.

Data stewardship and data ownership are two important concepts in defining, assessing / evaluating, improving, and controlling data quality. **Data owners** are individuals or groups who are responsible for the specific data content and its use as a data source, i.e., those who collect them, and use for their daily business activities (e.g., risk analysis or risk management). They can be employee from the commercial area, e.g., customer consultants, who are the owners of customer contact data. **Data stewards**, on the other hand, are less interested in content than in data structure. By analysing and documenting these structures and controlling the

implementation of data governance policies, they act as data quality monitors, e.g., bank risk management in banking. This systematic work of documenting and controlling technical requirements and deliverables assists IT departments in developing appropriate architectures for technical data quality protection. In the area of data management, data stewards are an important link between IT and business.

In a data management environment, data governance becomes a key component. Data governance as part of data management determines *what data needs to be defined more precisely, accurately or detailed, how it is expected to be done, how business expectations/requirements are taken into account and dealt with, and how data generally (can) change*. As a critical and decisive factor, data governance links business policy to data management and forms the regulatory framework / backbone when it comes to the systematic integration of data. Larger companies can afford to create their own data management team to complement the data governance committee with the development, governance, and implementation of data management tasks. In smaller companies, assigning roles as a secondary organization using existing processes and organizational structures is a more cost-effective alternative.

VI. CONCLUSIONS

Companies seeking to transform data into the value should be able to understand, process and manage data and data-centred artifacts such as algorithms efficiently and intelligently, paying attention to data quality, ethics and security and privacy concerns, being compliant with regulatory standards. However, most businesses face challenges in adapting to this reality, especially when adopting AI applications [9].

This paper introduced a framework for a data governance workflow for handling duplicate data. Successful data governance requires a proven strategy: a combination of governance framework and engaging the right people at all levels is critical. In today's digital and digitized corporate world, data are omnipresent and ubiquitous: they form the basis for organizational workflows, processes and decisions. In this regard, the adoption of an effective data governance framework is critical. This results in company employees using only accurate, unique, reliable, valid, trustworthy, useful, and valuable data.

In general, not only technological solutions are needed that would identify / detect poor quality data and allow their examination and correction, or would ensure their prevention by integrating some controls into the system design, striving for "data quality by design" [44-45], but also cultural changes related to data management and governance within the organization. These two perspectives form the basis of the wealth business data ecosystem. Thus, the presented framework describes the hierarchy of people who are allowed to view and share data, rules for data collection, data privacy, data security standards, and channels through which data can be collected. Ultimately, this framework will help users be more consistent in data collection and data quality for reliable and accurate results of data-driven actions and activities.

REFERENCES

- [1] M. Spiekermann, S. Wenzel, and B. Otto, "A Conceptual Model of Benchmarking Data and its Implications for Data Mapping in the Data Economy," in Multikonferenz Wirtschaftsinformatik 2018, Lüneburg, Germany, Mar. 6-9, 2018.

- [2] T. Redman, "To Improve Data Quality, Start at the Source," Harvard Business Review, 2020.
- [3] A. Gabernet and J. Limburn, "Breaking the 80/20 rule: How data catalogs transform data scientists' productivity," IBM, 2017. [Online]. Available: <https://www.ibm.com/blogs/bluemix/2017/08/ibm-data-catalog-data-scientists-productivity/>
- [4] A. Nikiforova, "Open Data Quality Evaluation: A Comparative Analysis of Open Data in Latvia," Baltic Journal of Modern Computing, vol. 6, no. 4, pp. 363–386, 2018.
- [5] J. E. Ross, Total quality management: Text, cases, and readings. Routledge, 2017.
- [6] A. Scriffignano, "Understanding Challenges and Opportunities in Data Management," Dun & Bradstreet, 2019, available online at <https://www.dnb.co.uk/perspectives/master-data/data-management-report.html>.
- [7] M. Chien and A. Jain, "Gartner Magic Quadrant for Data Quality Solutions," 2020. [Online]. Available: <https://www.gartner.com/en/documents/3988016/magic-quadrant-for-data-quality-solutions>.
- [8] P. C. Sharma, S. Bansal, R. Raja, P. M. Thwe, M. M. Htay, and S. S. Hlaing, "Concepts, strategies, and challenges of data deduplication," in Data Deduplication Approaches, T. T. Thwel and G. R. Sinha, Eds., Academic Press, 2021, pp. 37–55.
- [9] Strategy & part of the PwC Network. Chief Data Officer Study . online: https://www.strategyand.pwc.com/de/en/functions/data-strategy/cdo-2022.html?trk=feed-main-feed-card_feed-article-content, last accessed: 26/05/2023
- [10] N. Nataliia, H. Yevgen, K. Artem, H. Iryna, Z. Bohdan, and Z. Iryna, "Software System for Processing and Visualization of Big Data Arrays," in Advances in Computer Science for Engineering and Education. ICCSEEA 2022, Z. Hu, I. Dychka, S. Petoukhov, and M. He, Eds. Cham: Springer, 2022, vol. 134, pp. 151–160.
- [11] S. Bansal and P. C. Sharma, "Classification criteria for data deduplication methods," in Data Deduplication Approaches, Tin Thein Thwel and G. R. Sinha, Eds. Academic Press, 2021, pp. 69–96.
- [12] O. Azeroual, M. Jha, A. Nikiforova, K. Sha, M. Alsmirat, and S. Jha, "A Record Linkage-Based Data Deduplication Framework with DataCleaner Extension," Multimodal Technologies and Interaction, vol. 6, no. 4, p. 27, 2022.
- [13] I. Heumann, A. MacKinney, and R. Buschmann, "Introduction: The issue of duplicates," The British Journal for the History of Science, vol. 55, pp. 257–278, 2022.
- [14] B. Engels, "Data governance as the enabler of the data economy," Intereconomics, vol. 54, pp. 216–222, 2019.
- [15] R. Abraham, J. Schneider, and J. vom Brocke, "Data governance: A conceptual framework, structured review, and research agenda," International Journal of Information Management, vol. 49, pp. 424–438, 2019.
- [16] M. Fadler, H. Lefebvre, & C. Legner, "Data governance: from master data quality to data monetization," In ECIS, 2021.
- [17] A. Gregory, "Data governance — Protecting and unleashing the value of your customer data assets," J Direct Data Digit Mark Pract, vol. 12, pp. 230–248, 2011.
- [18] D. Che, M. Safran, and Z. Peng, "From Big Data to Big Data Mining: Challenges, Issues, and Opportunities," in Database Systems for Advanced Applications. DASFAA 2013, Hong et al., Eds. Springer, Berlin, Heidelberg, 2013, vol. 7827.
- [19] A. Donaldson and P. Walker, "Information governance—A view from the NHS," International Journal of Medical Informatics, vol. 73, pp. 281–284, 2004.
- [20] P. P. Tallon, R. V. Ramirez, and J. E. Short, "The information artifact in IT governance: Toward a theory of information governance," Journal of Management Information Systems, vol. 30, pp. 141–177, 2014.
- [21] K. Hildebrandt, F. Panse, N. Wilcke, and N. Ritter, "Large-Scale Data Pollution with Apache Spark," IEEE Transactions on Big Data, vol. 6, pp. 396–411, 2020.
- [22] Splunk, "What Is Dark Data?", 2019, online: https://www.splunk.com/en_us/data-insider/what-is-dark-data.html, last accessed: 10.05.2023
- [23] D. J. Grimm, "The dark data quandary," Am. UL Rev., vol. 68, pp. 761, 2018.
- [24] P. N. Edwards, M. S. Mayernik, A. L. Batcheller, G. C. Bowker, and C. L. Borgman, "Science friction: Data, metadata, and collaboration," Social studies of science, vol. 41, no. 5, pp. 667–690, 2011.
- [25] G. Gimpel, "Dark data: the invisible resource that can drive performance now," Journal of Business Strategy, vol. 42, no. 4, pp. 223–232, 2021.
- [26] D. Firmani, M. Mecella, M. Scannapieco, and C. Batini, "On the meaningfulness of 'big data quality'," Data Science and Engineering, vol. 1, pp. 6–20, 2016.
- [27] P. H. Verburg, K. Neumann, and L. Nol, "Challenges in using land use and land cover data for global change studies," Global change biology, 17(2), 974–989, 2011.
- [28] S. Sarawagi and A. Bhamidipaty, "Interactive deduplication using active learning," in Proceedings of the eighth ACM SIGKDD international conference on Knowledge discovery and data mining, 2002, pp. 269–278.
- [29] T. F. Kusumasari, "Data profiling for data quality improvement with OpenRefine," in 2016 international conference on information technology systems and innovation (ICITSI), IEEE, 2016, pp. 1–6.
- [30] O. Azeroual, G. Saake, and M. Abuosba, "Data Quality Measures and Data Cleansing for Research Information Systems," Journal of Digital Information Management, vol. 16, pp. 12–21, 2018.
- [31] G. Papadakis, E. Ioannou, C. Niederée, and P. Fankhauser, "Efficient entity resolution for large heterogeneous information spaces," in Proceedings of the fourth ACM international conference on Web search and data mining, 2011, pp. 535–544.
- [32] M. K. Alnoory and M. M. Aqel, "Performance evaluation of similarity functions for duplicate record detection," Middle East University, 2011.
- [33] S. R. Seaman and I. R. White, "Review of inverse probability weighting for dealing with missing data," Statistical methods in medical research, vol. 22, no. 3, pp. 278–295, 2013.
- [34] D. Nadeau and S. Sekine, "A survey of named entity recognition and classification," Lingvisticae Investigationes, vol. 30, no. 1, pp. 3–26, 2007.
- [35] C. Batini and M. Scannapieca, "Object Identification," in Data Quality: Concepts, Methodologies and Techniques, 2006, pp. 97–132.
- [36] F. Panse, M. van Keulen, and N. Ritter, "Indeterministic handling of uncertain decisions in deduplication," Journal of Data and Information Quality (JDIQ), vol. 4, no. 2, pp. 1–25, 2013.
- [37] F. Panse, M. van Keulen, and N. Ritter, "Indeterministic handling of uncertain decisions in duplicate detection," CTIT Technical Report Series, TR-CTIT-10-21, 2010.
- [38] F. Naumann, A. Bilke, J. Bleiholder, and M. Weis, "Data fusion in three steps: Resolving inconsistencies at schema-, tuple-, and value-level," IEEE Data Engineering Bulletin, vol. 29, no. 2, pp. 21–31, 2006.
- [39] O. Hassanzadeh and R. J. Miller, "Creating probabilistic databases from duplicated data," The VLDB Journal, vol. 18, no. 5, pp. 1141, 2009.
- [40] D. M. Powers, "Evaluation: from precision, recall and F-measure to ROC, informedness, markedness and correlation," arXiv preprint arXiv:2010.16061, 2020.
- [41] N. Wu, N., E. M. Pierce, J. R. Talburt & Wang, "An Information Theoretic Approach to Information Quality Metric," In ICIQ (pp. 133–145), 2006.
- [42] O. Azeroual, "Untersuchung der Datenqualität in FIS," in Untersuchungen zur Datenqualität und Nutzerakzeptanz von Forschungsinformationssystemen, Springer Vieweg, Wiesbaden, 2022.
- [43] A. Nikiforova. Open Data Quality. In 13th International Baltic Conference on Databases and Information Systems (DBIS 2018), Trakai, Lithuania, July 1–4, 2018, S. 151–160. Springer, Cham, 2018.
- [44] C., Guerra-García, A. Nikiforova, S. Jiménez, H. Perez-Gonzalez, M. Ramirez-Torres & L. Ontañón-García, ISO/IEC 25012-based methodology for managing data quality requirements in the development of information systems: Towards Data Quality by Design." Data & Knowledge Engineering 145 (2023): 102152.
- [45] D. C. Corrales, A. Ledezma, & J.C. Corrales, A systematic review of data quality issues in knowledge discovery tasks. Revista Ingenierías Universidad de Medellín, 15(28), 125–150, 2016.